

Many yeast species of the genus *Saccharomyces* are found throughout the world in either wild or domesticated populations. The domesticated populations are used by humans for industrial purposes (eg, food processing, pharmaceutical production). Studies have shown that different *Saccharomyces* species, which may reproduce either asexually or sexually, have hybridized over time and that some of these hybridizations have resulted in the formation of new species.

Studies of North American *Saccharomyces* yeast identified two new species thought to be the result of hybridization events. One species (*Sp3*) is thought to have arisen approximately 10,000 years ago from the hybridization of two parental species (*Sp1* and *Sp2*). Another hybrid yeast species (*Sp4*) is thought to be the result of a more recent hybridization event between *Sp1* and *Sp3*.

In a study of reproduction among these four yeast species, scientists performed multiple sets of crosses among various species. All sets of crosses produced a similar number of spores. Next, they determined the percentage of resulting spores (ie, cells capable of dividing and developing into mature yeast cells) from each set of crosses that was viable. The results are recorded in Table 1.

Table 1 Viability of Spores Resulting From Crosses Among Various *Saccharomyces* Yeast Species

Species crossed	Spore viability (%)
<i>Sp1</i> × <i>Sp1</i>	62.1
<i>Sp2</i> × <i>Sp2</i>	79.6
<i>Sp3</i> × <i>Sp3</i>	87.1
<i>Sp4</i> × <i>Sp4</i>	89.5
<i>Sp1</i> × <i>Sp4</i>	29.2
<i>Sp1</i> × <i>Sp2</i>	38.0
<i>Sp1</i> × <i>Sp3</i>	42.4
<i>Sp2</i> × <i>Sp3</i>	48.6
<i>Sp2</i> × <i>Sp4</i>	30.2
<i>Sp3</i> × <i>Sp4</i>	69.0

In addition, the scientists confirmed and examined the growth (ie, asexual reproduction) of all four yeast species under several environmental conditions. In most conditions, *Sp3* exhibited a higher growth rate than *Sp2* and a lower growth rate than *Sp1* over time. However, when grown in the presence of mannose, *Sp3* exhibited a higher growth rate than both *Sp2* and *Sp1*. *Sp4* exhibited the lowest growth rate among the species in most conditions.

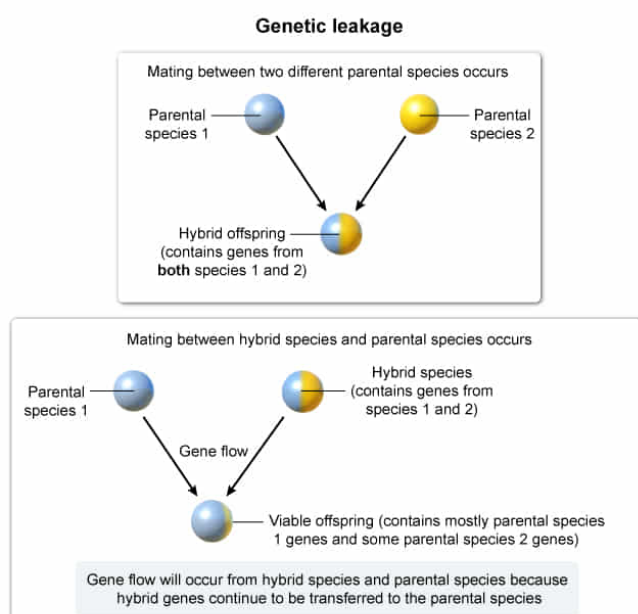
The scientists hypothesize that genetic leakage has occurred in some of the studied yeast lineages over time. Which of the following observations best supports this hypothesis?

- ☐ A. Hybrid individuals express certain phenotypes that differ from those expressed in the parental species. [50%]
- ☐ B. Hybrid individuals exhibit reproductive isolation with some other *Saccharomyces* species. [15%]
- ✓ ☒ C. Hybrid individuals can produce some viable offspring when mated with their parental species. [25%]
- ☐ D. Hybrid individuals frequently pass genetic information to clones of themselves via asexual reproduction. [8%]

Correct

25% answered correctly

Explanation:



When two individuals from two different parental species reproduce, their offspring are known as **hybrid** organisms. If these hybrid offspring successfully mate with members of either parental species, **genetic leakage** may occur. Genetic leakage is generally defined as the transfer of genetic information (ie, **genes**) between different species.

For example, genetic leakage occurs when an individual from a hybrid species and another from the parental species that formed the hybrid mate and produce viable offspring. When this mating takes place, offspring inherit genes from the parental species and from the hybrid species, which contains genes from two different parental species. If this type of mating continues over time, **gene flow** will occur *from* the hybrid species *to* the parental species with which it breeds because hybrid genes will continue to be transferred to the parental species.

The passage describes different *Saccharomyces* species that have mated with one another to form new hybrid species. According to the question, scientists hypothesize that genetic leakage has occurred in some of these studied lineages over time. Therefore, the observation that best supports the scientists' hypothesis is that **hybrid** individuals are able to produce some **viable offspring** when mated with individuals in their **parental species**.

(Choice A) Hybrid organisms may express phenotypes (ie, traits) that differ from those expressed in their parental species. However, genetic leakage occurs only when these hybrids are able to produce viable offspring with one of their parental species.

(Choice B) When two different species are *unable* to produce viable or fertile offspring from mating with each another, they are said to be **reproductively isolated**. Genetic leakage relies on the formation of *fertile* hybrids that *can* mate and produce viable offspring with one of their parental species. Therefore, the occurrence of reproductive isolation between hybrids and other *Saccharomyces* species would *not* support the hypothesis.

(Choice D) Frequent production of clones via **asexual reproduction** would allow hybrids to continuously pass their genes on to *identical* offspring but would *not* allow for the transfer of genes *between* hybrids and one of their parental species via genetic leakage.

Educational objective:

Hybrid offspring are produced when members of two different species mate. If these hybrid offspring mate with either parental species that produced the hybrid, genetic leakage (ie, the transfer of genes between different species) may occur.

Time spent:0 sec

Subject:

Biology

QID:403245

System:

Genetics and Evolution

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Studies of North American *Saccharomyces* yeast identified two new species thought to be the result of hybridization events. One species (*Sp3*) is thought to have arisen approximately 10,000 years ago from the hybridization of two parental species (*Sp1* and *Sp2*). Another hybrid yeast species (*Sp4*) is thought to be the result of a more recent hybridization event between *Sp1* and *Sp3*.

In a study of reproduction among these four yeast species, scientists performed multiple sets of crosses among various species. All sets of crosses produced a similar number of spores. Next, they determined the percentage of resulting spores (ie, cells capable of dividing and developing into mature yeast cells) from each set of crosses that was viable. The results are recorded in Table 1.

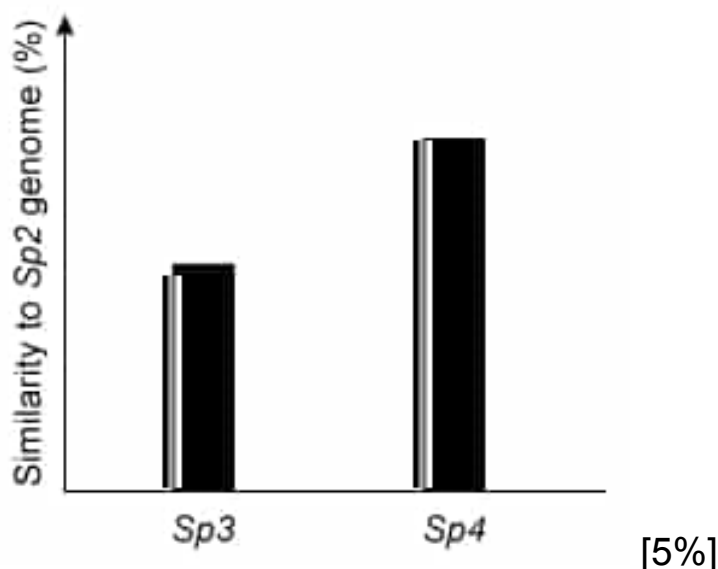
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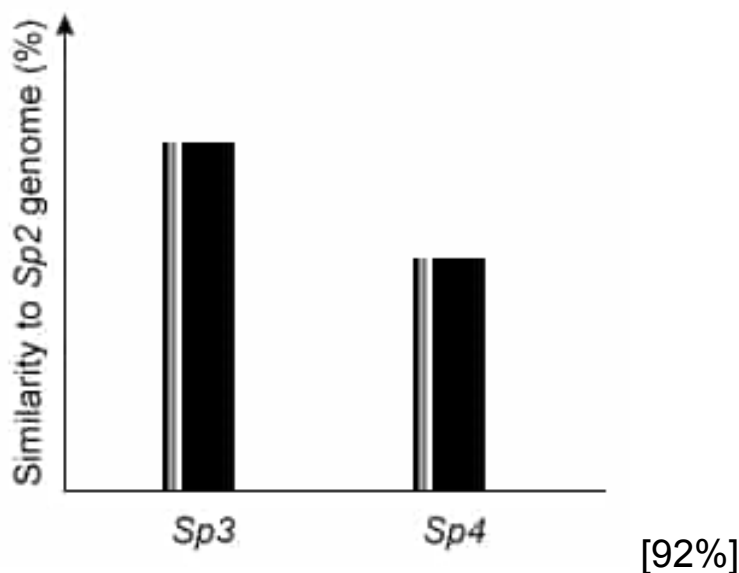
In addition, the scientists confirmed and examined the growth (ie, asexual reproduction) of all four yeast species under several environmental conditions. In most conditions, *Sp3* exhibited a higher growth rate than *Sp2* and a lower growth rate than *Sp1* over time. However, when grown in the presence of mannose, *Sp3* exhibited a higher growth rate than both *Sp2* and *Sp1*. *Sp4* exhibited the lowest growth rate among the species in most conditions.

The scientists sequence the genomes of all *Saccharomyces* yeast species included in the study. Based on the passage, which of the following graphs shows the expected levels of similarity between the genome of *Sp2* and the genomes of *Sp3* and *Sp4*?

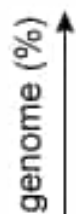
☐ A.

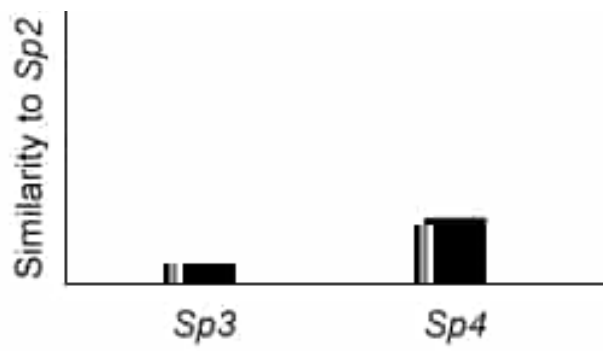


☒ B.



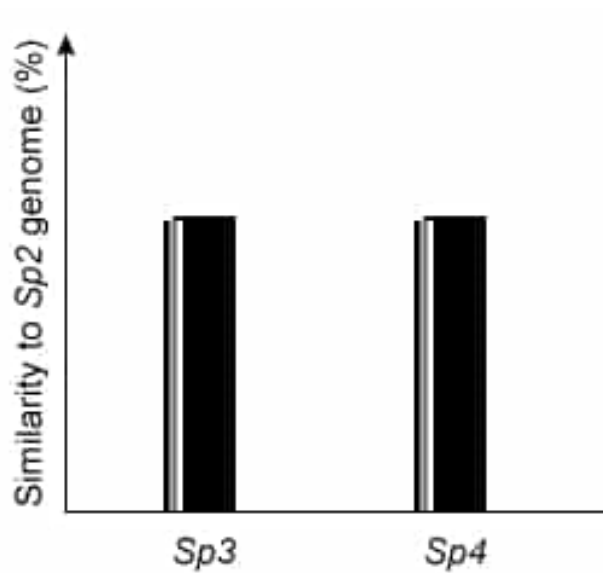
☐ C.





[1%]

○ D.



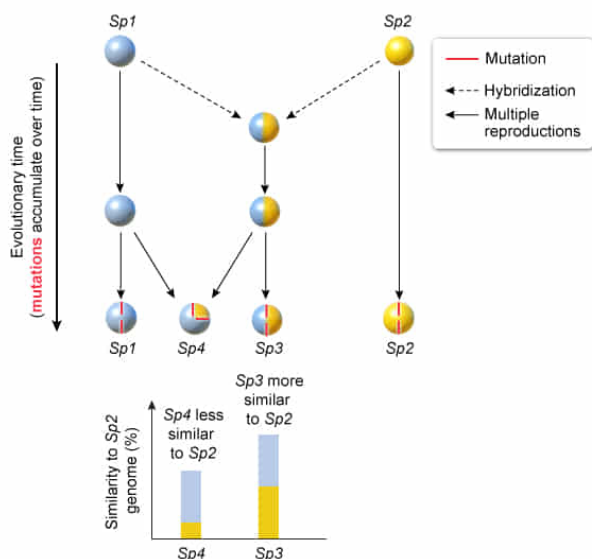
[1%]

Correct

92% answered correctly

Explanation:

Evolutionary time and genomic similarities among yeast species



Species generally accumulate neutral **mutations** at a fairly constant rate over evolutionary time. These mutation rates serve as a **molecular clock**, allowing researchers to measure evolutionary time and estimate **evolutionary relationships** among species. According to this model, species that are *more* closely related have had *less* time to accumulate different mutations and share *more* **genomic** similarities.

The passage states that hybridization events (ie, mating between two different species) among *Saccharomyces* yeast species have resulted in the formation of two new species. Specifically, the yeast species *Sp3* is thought to have arisen approximately 10,000 years ago from hybridization of parental species *Sp1* and *Sp2*. However, *Sp4* is thought to have arisen from a more recent hybridization event between *Sp1* and *Sp3*.

Accordingly, *Sp3* would share a common ancestor (and genomic similarities) with *Sp2* because *Sp3* was formed directly from hybridization between *Sp1* and *Sp2*. Over evolutionary time, however, neutral mutations have likely accumulated in the *Sp3* genome. Therefore, the more recent hybridization of *Sp3* with *Sp1* to produce *Sp4* would have resulted in *Sp4* hybrids sharing even *fewer* similarities with *Sp2* individuals.

Accordingly, *Sp3* individuals that descended *directly* from hybridization involving *Sp2* would likely share **more** genomic similarities with *Sp2* than *Sp4* individuals would share with *Sp2*. Only **Choice B** shows this result.

(Choices A, C, and D) Because *Sp3* originated directly from hybridization of *Sp1* and *Sp2*, there are likely more genomic similarities between *Sp2* and *Sp3* than between *Sp2* and *Sp4*, which was formed from hybridization between *Sp1* and *Sp3*.

Educational objective:

According to the molecular clock model, neutral mutations (ie, those that do not affect fitness) accumulate in an organism's genome at a fairly constant rate over evolutionary time. Species that are more closely related exhibit higher levels of genomic similarity than species that are less closely related.

Time spent:0 sec

Subject:
Biology

QID:403246

System:
Genetics and Evolution

Many yeast species of the genus *Saccharomyces* are found throughout the world in either wild or domesticated populations. The domesticated populations are used by humans for industrial purposes (eg, food processing, pharmaceutical production). Studies have shown that different *Saccharomyces* species, which may reproduce either asexually or sexually, have hybridized over time and that some of these hybridizations have resulted in the formation of new species.

Studies of North American *Saccharomyces* yeast identified two new species thought to be the result of hybridization events. One species (*Sp3*) is thought to have arisen approximately 10,000 years ago from the hybridization of two parental species (*Sp1* and *Sp2*). Another hybrid yeast species (*Sp4*) is thought to be the result of a more recent hybridization event between *Sp1* and *Sp3*.

In a study of reproduction among these four yeast species, scientists performed multiple sets of crosses among various species. All sets of crosses produced a similar number of spores. Next, they determined the percentage of resulting spores (ie, cells capable of dividing and developing into mature yeast cells) from each set of crosses that was viable. The results are recorded in Table 1.

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<i>Sp3</i> × <i>Sp4</i>	69.0

In addition, the scientists confirmed and examined the growth (ie, asexual reproduction) of all four yeast species under several environmental conditions. In most conditions, *Sp3* exhibited a higher growth rate than *Sp2* and a lower growth rate than *Sp1* over time. However, when grown in the presence of mannose, *Sp3* exhibited a higher growth rate than both *Sp2* and *Sp1*. *Sp4* exhibited the lowest growth rate among the species in most conditions.

Compared to crossing *Sp1* individuals with other *Sp1* individuals, crossing *Sp1* individuals with *Sp4* individuals (Table 1) resulted in:

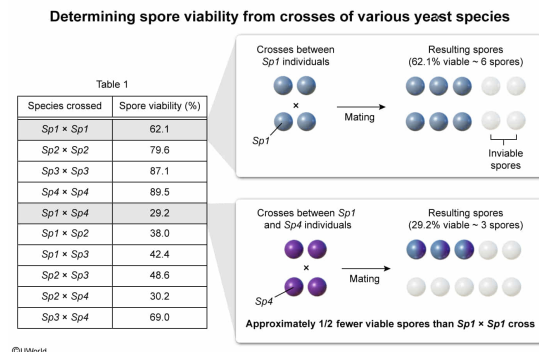
- ☐ A. approximately one-third fewer viable spores. [9%]

- ☐ B. approximately three times more viable spores.
- ✓ ☒ C. approximately one-half fewer viable spores. [88%]
- ☐ D. approximately two times more viable spores. [1%]

Correct

88% answered correctly

Explanation:



Mating between two different species may result in hybrid offspring, some of which may be inviable (ie, unable to survive) or infertile. When two different species mate and cannot produce viable or fertile offspring, they are considered to be **reproductively isolated** from each another.

There are two forms of reproductive isolation:

- **Prezygotic reproductive isolation** occurs when **fertilization** is *prevented* between the sperm of a male from one species and the egg of a female from another species (ie, occurs *before* zygote formation).
- **Postzygotic reproductive isolation** occurs when fertilization takes place but resulting offspring are **inviable** or infertile (ie, occurs *after* zygote formation).

The passage states that the scientists performed multiple crosses among various yeast species and determined the percentage of resulting spores (ie, cells capable of dividing and developing into mature yeast cells) that were viable from each set of crosses. All sets of crosses produced a *similar* number of spores.

In this question, the results in Table 1 can be used to compare spore viability of Sp1 × Sp1 crosses to that of Sp1 × Sp4 crosses (ie, Sp1 and Sp4 hybridization). According to Table 1, **62.1%** of spores were viable in Sp1 × Sp1 crosses. In contrast, only **29.2%** of spores were viable in Sp1 × Sp4 crosses. The approximate value by which these two percentages differ can be calculated as follows:

$$\text{Factor of difference between spore viabilities} = \frac{\text{Spore viability from Sp1} \times \text{Sp4 crosses}}{\text{Spore viability from Sp1} \times \text{Sp1 crosses}} = \frac{29.2}{62.1} \approx 0.5 \text{ or } \frac{1}{2}$$

Therefore, compared to Sp1 × Sp1 crosses, Sp1 × Sp4 crosses resulted in approximately **one-half fewer viable spores**. Although Sp1 × Sp1 crosses still resulted in some inviable spores, the greatly *reduced* spore viability resulting from hybridization between Sp1 and Sp4 provides evidence that *some* reproductive isolation is likely occurring between these two species.

(Choices A, B, and D) These values result from *incorrect* calculations.

Educational objective:

When two species cannot interbreed to produce viable or fertile offspring, they are said to be reproductively isolated from each another. Reproductive isolation may be prezygotic (before zygote formation) or postzygotic (after zygote formation).

Time spent:0 sec**Subject:**
Biology**QID:**403247**System:**
Genetics and Evolution

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In a study of reproduction among these four yeast species, scientists performed multiple sets of crosses among various species. All sets of crosses produced a similar number of spores. Next, they determined the percentage of resulting spores (ie, cells capable of dividing and developing into mature yeast cells) from each set of crosses that was viable. The results are recorded in Table 1.

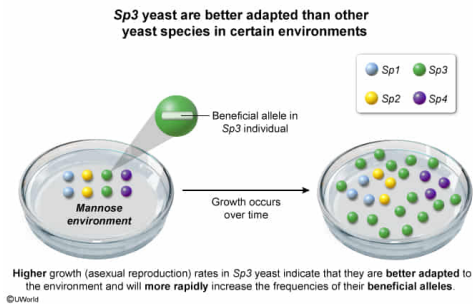
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In addition, the scientists confirmed and examined the growth (ie, asexual reproduction) of all four yeast species under several environmental conditions. In most conditions, *Sp3* exhibited a higher growth rate than *Sp2* and a lower growth rate than *Sp1* over time. However, when grown in the presence of mannose, *Sp3* exhibited a higher growth rate than both *Sp2* and *Sp1*. *Sp4* exhibited the lowest growth rate among the species in most conditions.

Based on the passage, which of the following statements regarding the *Saccharomyces* yeast species *Sp3* is most likely to be true?

- ☐ A. *Sp3* yeast are able to outcompete both of their parental species in most environments due to inheriting a hybrid genome. [12%]
- ☒ B. *Sp3* yeast propagate more effectively and increase the frequencies of their beneficial alleles more rapidly in some environmental conditions than other yeast species. [65%]
- ☐ C. *Sp3* yeast are the only species that can use mannose effectively as a source of energy during metabolism. [19%]
- ☐ D. *Sp3* yeast have very low genetic diversity levels because certain traits beneficial for growth were selected for throughout their evolution. [2%]

Explanation:

Natural selection is an **evolutionary mechanism** in which **adaptive traits** that increase the **fitness** (reproductive success) of an organism are *more* likely to be passed to the next generation than *less* favorable traits. This causes beneficial traits (those encoded by adaptive **alleles**) to become *more* common and deleterious traits to become *less* common over time.

In the study described in the passage, scientists confirmed and examined growth (ie, **asexual reproduction**) of various yeast species under several environmental conditions. They found that, in *most* conditions, Sp3 exhibited a *higher* growth rate than Sp2 and a *lower* growth rate than Sp1. However, Sp3 exhibited a *higher* growth rate than *both* Sp1 and Sp2 when grown in the presence of mannose (a type of sugar).

These findings indicate that **Sp3 yeast** have traits that are *more adaptive* (beneficial) in certain environments, allowing them to **propagate** (ie, grow and reproduce) **more effectively** than other yeast species in these conditions. Therefore, in **some** environmental conditions, Sp3 yeast are likely able to **increase the frequencies** of their **beneficial alleles more rapidly** than other species.

(Choice A) The passage states that, in *most* environmental conditions, Sp3 (formed from past hybridization of parental species Sp1 and Sp2) exhibited a *higher* growth rate than Sp2 and a *lower* growth rate than Sp1 over time. Therefore, Sp3 yeast are *not* able to outcompete *both* of their parental species in *most* conditions.

(Choice C) The passage states that scientists *confirmed* and examined growth of all species included in the study. Although Sp3 exhibited a higher growth rate than other yeast species when grown in a mannose-containing environment, confirmation of growth means that all species are able to grow in this environment.

(Choice D) Because Sp3 was able to exhibit growth in most environmental conditions, it is likely that this species has high levels of genetic diversity, which means that it possesses many alleles that allow it to survive in **different environments**. Therefore, it is *not* likely that Sp3 yeast have very low genetic diversity levels.

Educational objective:

During natural selection, beneficial traits (ie, those improving fitness) are more likely to be passed to subsequent generations and increase in frequency over time than less favorable traits.

Time spent: 0 sec
Subject:
Biology

QID: 403248
System:
Genetics and Evolution

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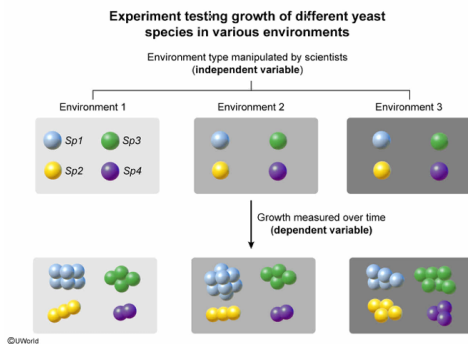
In the growth experiment conducted in the passage, which of the following best describes the independent variable?

- ☐ A. The different yeast species that were analyzed [35%]
- ☐ B. The growth rates of each yeast species over time [5%]
- ☐ C. The total number of yeast cells for each species initially included in the experiment [3%]
- ☒ D. The environmental conditions to which yeast species were exposed [56%]

Correct

55% answered correctly

Explanation:



In a scientific experiment, the **independent variable** is controlled (ie, manipulated) by the scientists and is *not* affected by other experimental variables. The *response* of a dependent variable to this independent variable can then be measured and analyzed.

In the experiment described in the passage, the scientists examined the growth (ie, **asexual reproduction**) of different yeast species in various environmental conditions. The scientists *manipulated* the environmental conditions to see how the growth rates of each species (ie, the dependent variable) would be affected (**Choice B**). Therefore, the **independent variable** in this experiment was the **environmental conditions to which yeast species were exposed**.

(Choice A) The analyzed yeast species (*Sp1*, *Sp2*, *Sp3*, *Sp4*) were *not* manipulated by the scientists throughout the experiment. In addition, these *same* four species were analyzed in each environmental condition. Therefore, the yeast species being analyzed was *not* the independent variable.

(Choice C) The total number of yeast cells for each species initially included in the experiment should be held *constant* (ie, serve as a **control variable**) throughout the experiment. This would ensure that growth rates *only* changed in response to differences in environmental conditions (the independent variable), *not* in response to differences in the initial total number of yeast cells.

Educational objective:

The independent variable is manipulated by scientists to determine how it affects a dependent (measured) variable.

Time spent: 0 sec

Subject:
Biology

QID: 403249

System:
Genetics and Evolution